

$$(AB)^T = B^T A^T$$

$$B^T = B$$

2. Dadas las Matrices:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, B = \begin{pmatrix} 0 & 3 \\ 3 & 5 \end{pmatrix}$$

Calcular:

a) A^T

b) B^T

c) $(AB)^T$

d) $A^T B^T$

e) $B^T A^T$ ←

$$\begin{pmatrix} 0 & 3 \\ 3 & 5 \end{pmatrix}$$

$$a) = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$$

$$b) \rightarrow \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 0 & 3 \\ 3 & 5 \end{pmatrix} = \begin{pmatrix} 1 \cdot 0 + 2 \cdot 3 & 1 \cdot 3 + 2 \cdot 5 \\ 3 \cdot 0 + 4 \cdot 3 & 3 \cdot 3 + 4 \cdot 5 \end{pmatrix} = \begin{pmatrix} 6 & 13 \\ 12 & 29 \end{pmatrix}^T$$

$$e) \rightarrow \begin{pmatrix} 0 & 3 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 0 \cdot 1 + 3 \cdot 2 & 0 \cdot 3 + 3 \cdot 4 \\ 3 \cdot 1 + 5 \cdot 2 & 3 \cdot 3 + 5 \cdot 4 \end{pmatrix} = \begin{pmatrix} 6 & 12 \\ 13 & 29 \end{pmatrix}$$

$$\Rightarrow \boxed{\begin{pmatrix} 6 & 12 \\ 13 & 29 \end{pmatrix}} = (AB)^T$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix}$$

A, B, C son matrices de $m \times n$

$$2(A+B) = C+B$$

$$2A + 2B = C + B \quad / -B$$

$$2A + 2B - B = C + B - B$$

$$\underline{2A + B = C}$$

3. Calcular la matriz X , si:

$$-2 \left(X + \overbrace{\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}}^A \right) = 3X + \overbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}}^B$$

$$-2(X+A) = 3X+B$$

$$-2X - 2A = 3X + B \quad / +2X$$

$$-2A = 5X + B \quad / -B$$

$$-2A - B = 5X \quad / \frac{1}{5}$$

$$\frac{1}{5}(-2A - B) = X$$

$$-\frac{1}{5}(2A+B) = X \quad \left| \quad X = -\frac{1}{5} \left(2 \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right) \right.$$

$$= -\frac{1}{5} \left(\begin{pmatrix} 2 & 4 & 6 \\ 0 & 2 & 4 \\ 0 & 0 & 2 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right)$$

$$\left| X = -\frac{1}{5} \begin{pmatrix} 3 & 4 & 6 \\ 0 & 2 & 4 \\ 0 & 0 & 3 \end{pmatrix} \right|$$

$$X = \begin{pmatrix} -\frac{3}{5} & -\frac{4}{5} & -\frac{6}{5} \\ 0 & -\frac{2}{5} & -\frac{4}{5} \\ 0 & 0 & -\frac{3}{5} \end{pmatrix}$$

$k \in \mathbb{R}$

$$k \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} ka & kb \\ kc & kd \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}; B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\rightarrow (-x-y) = -(x+y)$$

$$I_3 = I_3$$

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$O_2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

5. Muestre que la matriz:

$$A = \begin{pmatrix} 2 & 3y \\ \frac{3}{y} & 2 \end{pmatrix}$$

satisface la ecuación:

$$\rightarrow A^2 - 4A - 5I_2 = O_2$$

$$\frac{3}{y} \cdot \frac{2}{1} = \frac{6}{y}$$

$$A^2 = A \cdot A = \begin{pmatrix} 2 & 3y \\ \frac{3}{y} & 2 \end{pmatrix} \begin{pmatrix} 2 & 3y \\ \frac{3}{y} & 2 \end{pmatrix} = \begin{pmatrix} 4 + 3y \cdot \frac{3}{y} & 6y + 6y \\ \frac{6}{y} + \frac{6}{y} & 3y \cdot \frac{3}{y} + 4 \end{pmatrix} = \begin{pmatrix} 13 & 12y \\ \frac{12}{y} & 13 \end{pmatrix}$$

$$4A = 4 \begin{pmatrix} 2 & 3y \\ \frac{3}{y} & 2 \end{pmatrix} = \begin{pmatrix} 8 & 12y \\ \frac{12}{y} & 8 \end{pmatrix}$$

$$5I = 5 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}$$

$$A^2 - 4A - 5I = \begin{pmatrix} 13 & 12y \\ \frac{12}{y} & 13 \end{pmatrix} - \begin{pmatrix} 8 & 12y \\ \frac{12}{y} & 8 \end{pmatrix} - \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}$$

$$= \begin{pmatrix} 13 - 8 - 5 & \cancel{12y} - \cancel{12y} - 0 \\ \cancel{\frac{12}{y}} - \cancel{\frac{12}{y}} - 0 & 13 - 8 - 5 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$A \cdot 11 = A$$

$$A \cdot A^{-1} = A^{-1} \cdot A = 11$$

9. Encuentre la matriz X de tal forma que:

$$X \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$X \cdot A = 11 \quad | \cdot A^{-1} \rightarrow X \cdot \underbrace{A \cdot A^{-1}}_{11} = 11 \cdot A^{-1}$$

$$X \cdot 11 = A^{-1}$$

$$X = A^{-1}$$

$$X = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}^{-1}$$

$$x \cdot a = 1 \quad | \cdot \frac{1}{a}$$

$$\frac{x \cdot a}{a} = \frac{1}{a}$$

$$x = \frac{1}{a} \Rightarrow x = a^{-1}$$

$$X = \begin{pmatrix} -2 & 1 \\ -3/2 & -1/2 \end{pmatrix}$$

$$1 - 3 \cdot 0$$

$$0 - 3 \cdot 1 = -3$$

$$4 - 3 \cdot 2 = 4 - 6 = -2$$

$$3 - 3 \cdot 1 = 0$$

$$\begin{pmatrix} 1 & 2 & | & 1 & 0 \\ 3 & 4 & | & 0 & 1 \end{pmatrix} \quad F_2 = F_2 - 3F_1$$

$$\begin{pmatrix} 1 & 2 & | & 1 & 0 \\ 0 & -2 & | & -3 & 1 \end{pmatrix} \quad F_2 = -\frac{1}{2} F_2 \quad \begin{matrix} \cdot \frac{+1}{2} \\ \cdot \frac{+1}{2} \end{matrix}$$

$$\begin{pmatrix} 1 & 2 & | & 1 & 0 \\ 0 & 1 & | & 3/2 & -1/2 \end{pmatrix} \quad F_1 = F_1 - 2F_2$$

$$1 - 2 \cdot 0$$

$$2 - 2 \cdot 1 = 0$$

$$1 - 2 \cdot 3/2 = -2$$

$$0 - 2 \cdot (-1/2) = 1$$

$$\begin{pmatrix} 1 & 0 & | & -2 & 1 \\ 0 & 1 & | & 3/2 & -1/2 \end{pmatrix} \quad 11_2$$

$$A^{-1} = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix}$$